

2022

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers in
their own words as far as practicable.*

The figures in the margin indicate full marks.

*Answer any **five** questions in which*

Q. No. 1 is compulsory.

1. Answer any **four** questions of the following :

$$3\frac{1}{2} \times 4 = 14$$

- (a) Define Inverting and Non-inverting amplifiers.
- (b) What is balanced output differential amplifier ?
- (c) What are comparators ?
- (d) What do you mean by frequency response of an oscillator ?

- (e) Define DSBSC modulation.
- (f) What is velocity modulation ?
2. Discuss with circuit diagram or configuration a differential amplifier. 14
 3. Sketching the block diagram of an operational amplifier, discuss OP-AMP with negative feedback. 14
 4. Drawing the circuit diagram of a phase shift oscillator, discuss its functions and obtain the expression for its frequency response. 14
 5. Define Amplitude Modulation. Discuss the generation of AM waves. $6+8 = 14$
 6. Define multiplexer. Discuss, in detail, Frequency Division Multiplexing(FDM). $4+10 = 14$
 7. Define non-volatile memories. Discuss magnetic, optical and ferroelectric memories. 14
 8. Discuss with theory the working of Reflex Klystrons. 14

9. Write short notes on any **two** of the following :

7+7 = 14

- (a) Voltage follower
- (b) Adders
- (c) Vestigial side band modulator
- (d) Travelling Wave Tubes

